

Dynamic Deep Pixel Distribution Learning for Background Subtraction

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Abstract—Previous approaches to background subtraction usually approximate the distribution of pixels with artificial models. In this paper, we focus on automatically learning the distribution, using a novel background subtraction model named Dynamic Deep Pixel Distribution Learning (D-DPDL). In our D-DPDL model, a distribution descriptor named Random Permutation of Temporal Pixels (RPoTP) is dynamically generated as the input to a convolutional neural network for learning the statistical distribution, and a Bayesian refinement model is tailored to handle the random noise introduced by the random permutation. Because the temporal pixels are randomly permuted to guarantee that only statistical information is retained in RPoTP features, the network is forced to learn the pixel distribution. Moreover, since the noise is random, the Bayesian theorem is naturally selected to propose an empirical model as a compensation based on the similarity between pixels. Evaluations using standard benchmark demonstrates the superiority of the proposed approach compared with the state-of-the-art, including traditional methods as well as deep learning methods.

Index Terms—Background subtraction, Motion Detection, Deep Learning, Pixel Distribution, Random Permutation

I. INTRODUCTION

Motion is a powerful cue for image and scene segmentation in the human visual system. The human ability to detect, segment and understand motion is nearly instantaneous and works for diverse complex scenes. While there has been considerable recent progress related to understanding motion, it still appears that we are far from reaching human capabilities. Therefore, as an essential and fundamental research topic related to motion in computer vision, background subtraction has been attracting increasing attention. Especially, with the utilization of deep learning networks, several researches have recently achieved excellent performance. However, such sophisticated algorithms related to deep learning usually require hundreds of labelled frames for training, which is too expensive to be acceptable in real applications, since every pixel has to be labelled. Therefore, a general solution to background subtraction remains a challenging problem given limited labelled frames.

Essentially, background subtraction is a classification of pixels in a sequence of image frames, typically assuming a static camera, wherein every pixel of a frame is classified as foreground or background by comparing its current measurement with historical observations. This is a challenging problem when dealing with diverse and complex scenes, since

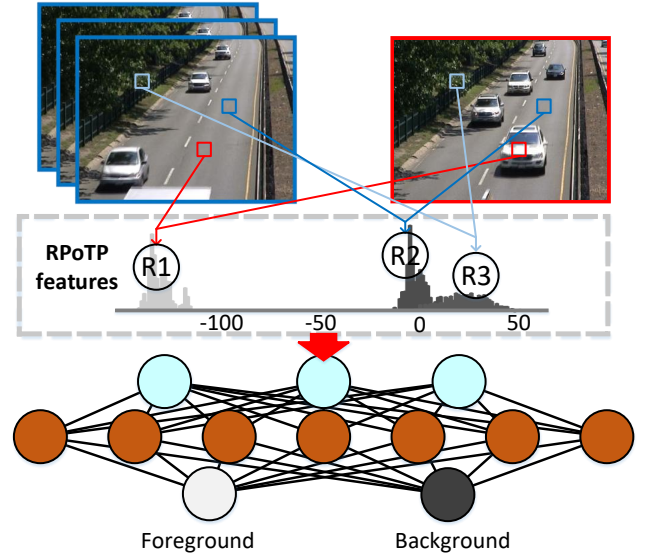


Fig. 1. An illustration of the deep pixel distribution learning model. The RPoTP features encode the distributions of pixel observations that belong to moving objects R1, static background R2 and dynamic background R3 respectively. The RPoTP features from all pixels are then fed into a convolutional neural network to learn a classifier for background subtraction.

the pixel measurements take the form of complex distributions. For example, as shown in R1, R2 and R3 of Fig. 1, the distributions of the pixel observations in the moving objects, the static and the dynamic backgrounds are completely different, and manually-tailored models have limited ability to cope with these contrasting distributions. To handle this problem, we focus on learning the distribution automatically instead of devising a sophisticated model. A deep learning network is naturally selected to learn the distribution, given its excellent learning ability. Then, a novel background subtraction model named Dynamically Deep Pixel Distribution Learning (D-DPDL) that needs limited groundtruth frames is proposed.

In our D-DPDL model, the distribution of pixels' observations is described by the Random Permutation of Temporal Pixels (RPoTP) feature, in which the pixel observations are randomly permuted to guarantee that only the statistical information is retained. The RPoTP features are dynamically generated as the input of the convolutional neural network (CNN) for every training epoch. This indirectly forces the network to rely solely on the statistics of the pixel intensity distribution. However, the utilization of random permutations unavoidably introduces random noise, and a Bayesian refine-

ment model based on similarity between pixels is proposed to compensate for the random noise and improve the accuracy of the proposed approach. Since each RPoTP feature only depends on a single pixel, a large number of features are obtained for training, even with limited groundtruth frames. Moreover, the proposed approach is valid even under the condition that training frames and testing frames are obtained from different scenes, because the distribution information is a general feature regardless of the scene. The main differences between this paper and our previous work [1] are:

- 1) We improve the procedure of distribution learning, in which the Random Permutation of Temporal Pixels (RPoTP) features are dynamically generated during the training to prevent the network from over fitting the pattern implied in random permutations.
- 2) We propose a Bayesian refinement model to compensate for the random noise generated from the utilization of random permutations in the extraction of the RPoTP features for training the network. The accuracy of the proposed approach is improved after compensating with the Bayesian refinement model, which is based on the assumption that similar neighboring pixels share similar labels of foreground or background.
- 3) Comprehensive experiments are proposed for evaluating the proposed approach, including *a)* the validation of the proposed approach under the condition that training videos and testing videos are different, as shown in Section IV-A, *b)* the effectiveness of the proposed Bayesian refinement model and strategy of dynamical RPoTP feature generation is shown in Section IV-B, *c)* a comprehensive comparison between the proposed approach and state-of-the-art methods including traditional and deep learning approaches is shown in Section IV-C.

II. RELATED WORK

In this section, we cover the progression of background subtraction algorithms over three sections. In Section II-A, traditional background subtraction methods are introduced. Background subtraction based on earlier machine learning techniques are discussed in Section II-B, while recent deep learning methods for background subtraction are described in Section II-C.

A. Traditional Algorithms

Traditionally background subtraction is done by modeling the variation of pixel observations over time (e.g. [2]–[19]). Among them, one of the most popular techniques for background subtraction [20] is the Gaussian mixture model (GMM) [4], where pixel distribution is approximated by several Gaussian functions. Numerous extensions of GMM have been proposed. For example, Varadarajan et al. [7] approximate the distribution from regions instead of pixels for background subtraction, where the neighboring relationships between pixels are also considered. Similarly, Sriram et al. [21] extend approximation through the use of expectation maximization. Moreover, several sophisticated algorithms utilize the kernel density estimation (e.g. [22]–[26]) instead of Gaussians for

approximation. Recently, Haines et al. [8], [27] proposed the use of Dirichlet processes with Gaussian mixture models to analyze pixel distribution, while Chen et al. [28], [29] used Gaussian mixture models to represent the vertices of spanning trees.

Besides GMMs, there are also several other recent advances (e.g., [15], [30]–[34]) in background subtraction. Charles et al. [15] utilized temporal binary features and color information for background subtraction. Moreover, inspired by the traditional codebook algorithm [35], they also utilized word dictionaries for background subtraction [34]. Zeng et al. [30] proposed the use of a histogram based on strong uniform fuzzy partitions, where the threshold for background segmentation is set adaptively according to the shape of the histogram. Sajid et al. [31] proposed a universal multi-modal system that merged pixels together to form what the authors called mega-pixels. Javed et al. [36] proposed a robust principal component analysis framework, which incorporates spatial and temporal sparse subspaces. Finally, Huynh et al. [33] subtracted the moving objects from a dynamic background by analyzing the motion of pixels captured from the comparison between current and background frames, while Jiang et al. [32] utilized a weighted-sample method to rapidly adapt to changing scenarios, and Yong et al. [18] proposed online matrix factorization for background subtraction.

However, these hand-crafted models do not work effectively when applied to a wide range of scene categories. For example, GMM-based algorithms suffer from performance degradation in scenes with high complexity [36], where the distributions of pixels are too complex to be described even by several Gaussian components. In contrast, our approach involves having a deep network adaptively learn the optimal classification mechanism from sample distributions of pixel intensities.

B. Algorithms based on Supervised Machine Learning

Besides traditional algorithms, there are also several background subtraction methods [37]–[47] that utilize machine learning, commonly involving support vector machines (SVM) and Bayesian methods. Lin et al. [37] proposed using a probabilistic SVM for background initialization, Cheng et al. [38] accommodated spatial interactions by minimizing a risk function generalization of the one class SVM, and Han et al. [39] utilized density-based features as the input to an SVM classifier. Zhang et al. [40], [41] proposed an imbalance compensation mechanism based on bilayer modeling and Bayesian classification. Culibrk et al. [42] constructed an unsupervised Bayesian classifier for background modeling and subtraction using a neural network architecture. The novel subspace learning method, proposed by Zhou et al. [43], extracted a sequence of regular video bricks. Then, it posed the background modeling problem as pursuing a subspace representation for video bricks while adapting to scene variations.

In addition to the above, there are also related works employing neural networks for background subtraction. Gregorio et al. [45] proposed a weightless neural network for dynamically adapting to background change, while Maddalena et al. [46], [47] proposed having background models be automatically generated through a self-organizing network, which

appears to work well in several scenes containing gradual illumination variation and camouflage.

C. Algorithms based on Deep Learning

As we all know, deep learning has become popular in diverse applications of computer vision, such as image classification [48]–[50], image segmentation [51], video captioning [52] and so on. Since background subtraction can be considered as a pixel-wise classification, or image segmentation based on spatial information, it is not surprising that recent background subtraction methods [15], [53]–[62] have embraced deep learning. Wang et al. [53] considered background subtraction as scene segmentation, and proposed a fully connected network with cross-entropy loss function to learn the background scenes directly. Similarly, Zeng et al. [58] proposed the multi-scale strategy to generate better results and Lim et al. [54] extracted multi-scale features of background scenes in the middle layers of a network. Moreover, Lim et al. [55] proposed the Feature Pooling Module, which consists of pooling kernels with different sizes and parameters, to alleviate the need for multiple scales. Zheng et al. [56] utilized Bayesian generative adversarial networks for background subtraction to improve the performance. Braham et al. [59] employed deep networks to evaluate the difference between current frames and a background image, the latter being computed as the temporal median across multiple frames.

In addition, a more robust background model algorithm was proposed by Babaee et al. [60] for background subtraction, with a network used for subtracting the background from the current image. Liang et al. [61] proposed a guided learning strategy to avoid manual labeling, in which the network is initiated with the foreground mask generated by the SubSENSE algorithm [15]. Zeng et al. [62] combined several background subtraction methods with the convolutional neural network. Although some of these excellent advances have achieved almost perfect results in standard datasets, many frames are needed in the training phase to achieve accurate segmentation of moving objects. However, since every pixel of a frame needs a label of foreground or background in background subtraction, the groundtruth masks of background subtraction are expensive, which limits the applications of these excellent methods.

Therefore, our proposed method departs from the conventional approach of creating an explicit representation of the background scene. Instead, we focus on the more fundamental concept underlying background subtraction, which is the classification of pixels in a time sequence [2], based on network-learned discriminative features that act directly on the distribution of pixels over time. Since a large quantity of distribution information can be generated with a limited number of groundtruth frames, the groundtruth data assumed by the proposed approach is much less than previous approaches. Moreover, the proposed approach is effective when training videos and testing videos are from different sources, which is demonstrated in Section IV-A, since distributions of temporal pixels have a considerable independence from scenes. As such, the distribution depends on a pixel itself even under the influence of interactions between neighbouring pixels.

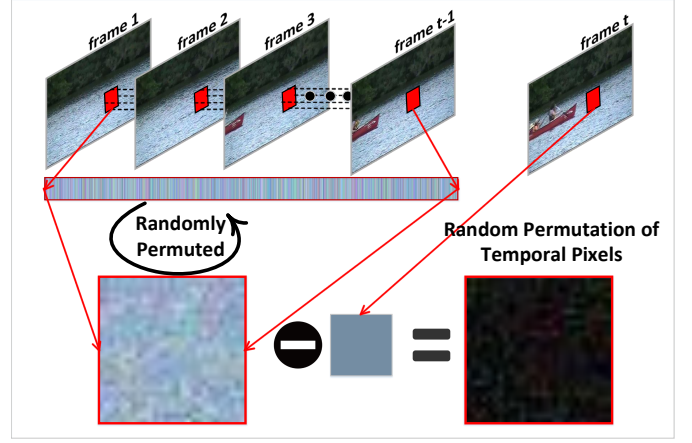


Fig. 2. The extraction of the Random Permutation of Temporal (RPoTP) pixels feature in a particular pixel. The historical observations of pixels are randomly permuted and reshaped into a rectangle. The RPoTP feature is then captured from the difference between the pixels in the rectangle and the current pixel.

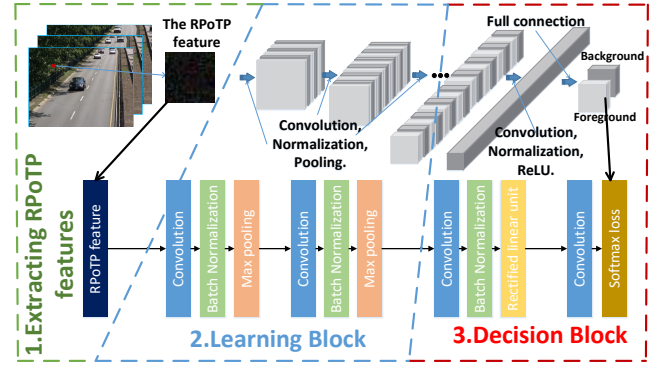


Fig. 3. Illustration of the deep pixel distribution learning model. The RPoTP features from all pixels are fed into the convolutional neural network to learn a classifier for background subtraction.

TABLE I
DETAILS OF OUR NETWORK ARCHITECTURE, WHICH CONSISTS OF 4 CONVOLUTIONAL LAYERS, 3 BATCH NORMALIZATION, 2 MAX POOLING AND A SOFTMAX OPERATOR.

Type	Filters	Layer size	Data size
The RPoTP feature			$25 \times 25 \times 3$
Convolution	64	$7 \times 7 \times 3$	$19 \times 19 \times 64$
Batch Normalization			$19 \times 19 \times 64$
Max pooling		4×4	$16 \times 16 \times 64$
Convolution	128	$7 \times 7 \times 64$	$10 \times 10 \times 128$
Batch Normalization			$10 \times 10 \times 128$
Max pooling		4×4	$7 \times 7 \times 128$
Convolution	1024	$7 \times 7 \times 128$	$1 \times 1 \times 1024$
Batch Normalization			$1 \times 1 \times 1024$
Rectified linear unit			$1 \times 1 \times 1024$
Convolution	255	$1 \times 1 \times 1024$	$1 \times 1 \times 255$
Softmax			

III. DYNAMIC DEEP PIXEL DISTRIBUTION LEARNING

The details of our Dynamic Deep Pixel Distribution Learning (D-DPDL) model are explained in this section. As shown in Fig. 4, the proposed approach is divided into two parts: “Pixel Distribution Learning,” in which the RPoTP features are dynamically generated as the input of a convolutional neural

network for learning the distribution of pixels (Section III-A), and the “Bayesian Refinement Model,” which is proposed to compensate for the random noise (Section III-B).

A. Deep Pixel Distribution Learning

For the completeness of this paper, we first introduce our previous Deep Pixel Distribution Learning (DPDL) [1] model. Then, the dynamic training strategy, which is one of the main differences between D-DPDL model and our previous DPDL, is discussed.

In videos obtained from natural scenes, multiple factors contribute to the variation of observations in a particular pixel, such as illumination change, dynamic background and moving objects. More importantly, the distribution of observations is different depending on the causes of the variation. For example, variations resulting from illumination changes typically follow a smooth unimodal distribution, since the position of the sun changes slowly during a time sequence. Conversely, for pixels in a dynamic background, the variation often takes the form of a multi-peak distribution, due to the pattern of cyclical repetitions in background pixels, such as leaves fluttering. However, when the variation is caused by moving objects, the observations are typically outliers compared to the distribution formed by the observations of pixels in the background. Therefore, the main purpose of the proposed approach is learning the distribution of pixels in different scenarios to classify these pixels into foreground or background. In particular, the Random Permutation of Temporal Pixels (RPoTP) feature is proposed to discriminate among distributions. The process of extracting the RPoTP features is shown in Fig. 2. An image sequence is denoted as $\{I_t | t \in [1, T]\}$ where T is the number of frames, captured with a static camera. The temporal observations for a particular pixel are obtained as a vector and reshaped into a square matrix, in which the observations are randomly permuted by a random permutation $\mathcal{X} = \{a_i | i \in [1, N]\}$ where N is the length of the permutation. The square matrix includes a fair sample set of historical observations, and the random permutation guarantees that only statistical information remains in the matrix. Finally, the RPoTP feature is extracted by subtracting the current pixel value from the sample matrix, in order to explicitly describe the difference from historical observations, which is mathematically shown as:

$$RPoTP_{x,y}^r(m, n) = I_t(x, y) - I_{\mathcal{X}(m \cdot r + n)}(x, y) \quad (1)$$

where $RPoTP_{x,y}^r$ denotes the RPoTP feature extracted from the pixel located at (x, y) , r is the user parameter to control the size of the RPoTP matrix, m, n are the indices of an entry in a matrix. During all the experiments, the radius r of the RPoTP feature is set to 25.

Essentially, background subtraction is a classification of pixels over time. Pixels are classified as foreground or background based on the comparison with their historical observations. The RPoTP features are actually the statistical representation of the comparison, which is used as the input to a convolutional neural network for classification. Therefore, the architecture of our network is devised for classification and the whole

classification procedure is shown in Fig. 3, which can be divided into three parts, features extraction, learning block and decision block. First, the RPoTP feature is extracted for each pixel and used as the input of the learning block. The learning block consists of several stacks of convolutional, normalization and pooling layers. The decision block is a set of convolutional, normalization and ReLU layers, followed by a fully connected layer that implements the softmax operator. Mathematically, it is shown as follows:

$$\ell_{x,y} = D(\mathcal{L}^\theta(RPoTP_{x,y}^r)), \quad (2)$$

where $\ell_{x,y}$ is a binary label of the pixel at location (x, y) identifying it as foreground or background, $\mathcal{L}$ is the learning block and $\mathcal{D}$ is the decision block. θ denotes the parameters of the learning block. The learning block $\mathcal{L}$ consists of several stacks of convolutional, normalization and pooling layers with θ determining the number of these layers. The decision block $\mathcal{D}$ is a set of convolutional, normalization and ReLU layers, followed by a fully connected layer that implements the softmax operator. During training, the logistic loss on the final output node is minimized.

For our previous DPDL model, the data input into the network for training is only generated once. Thus the input of the network is the same for every training epoch. Under this condition, it is possible that the network overfits to the pattern implied in random permutations rather than learns the statistical information included in RPoTP features. In order to address this issue, a dynamic training strategy is proposed as a compensation, in which the entries of RPoTP features are randomly re-permuted by new permutations for every training epoch. This strategy effectively prevents our network from overfitting, and improves the accuracy of the proposed approach in complex scenes. Section IV-A demonstrates the effectiveness of the dynamical training strategy by comparing our previous DPDL and the proposed D-DPDL.

In addition, there are also several minor modifications in the network architecture in the proposed D-DPDL model compared with our previous DPDL model. First, we increased the width of network and decreased the depth inspired by [63] which suggests wider networks tend to learn pixel distribution features. Besides, the sizes of convolutional kernels are increased as well, to allow more temporal pixels to be input into the network for learning distributions. Moreover, the stride of the pooling layer is set to 1 to avoid missing useful information during the down-sampling procedure. The comprehensive details of the network architecture are shown in Table I.

B. Bayesian Refinement Model

To the best of our knowledge, pixels are not independent but related to their neighborhoods, especially similar ones. It is common to accept that similar pixels should share similar labels of foreground or background. Such similarity is very useful against random noise. The Bayesian refinement model based on the similarity among pixels is thus proposed to compensate for random noise generated during distribution learning.

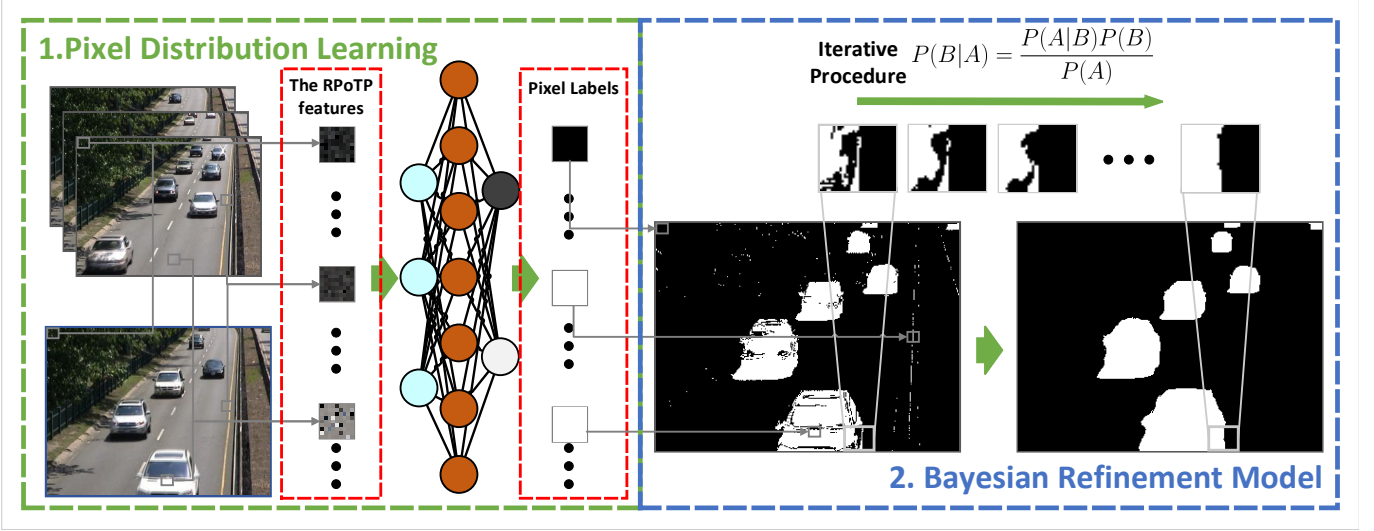


Fig. 4. The pipeline of the proposed approach, which is divided into two parts. The distribution included in the Random Permutation of Temporal Pixels (RPoTP) feature is learned by the convolutional neural network for background subtraction. The Bayesian refinement model is then proposed to compensate for the random noise introduced by random permutation.

The Bayesian refinement model is devised as an iterative procedure to infer the labels of pixels according to their neighborhoods, with the Bayesian theorem as the kernel of inference. As shown in the right part of Fig. 4, the foreground mask produced by the pixel distribution learning, which is discussed in Section III-A, is iteratively refined based on the Bayesian theorem. We explain the process of refinement in a pixel during one iteration, but the procedure is identical for every pixel during all iterations. Let us denote the features captured from the center pixel and their neighborhoods as v and $\mathcal{G}$ respectively. The procedure of inferring the label of the center pixel v is as follows:

$$P(a_i|v) = \frac{P(v|\mathcal{G})P(\mathcal{G}|a_i)P(a_i)}{P(v)}, \quad (3)$$

where $a_i \in \{0, 1\}$ denotes the labels of the pixels, which is either background represented by 0 or foreground represented by 1. $P(v)$ and $P(a_i)$ are the marginal probabilities corresponding to the feature and the label respectively. There is no need to consider $P(v)$ in the Bayesian inference procedure, since it is a constant. $P(v|\mathcal{G})$ and $P(\mathcal{G}|a_i)$ are the conditional probabilities, in which $P(\mathcal{G}|a_i)$ describes the probability of pixels with the label a_i belonging to $\mathcal{G}$, and $P(v|\mathcal{G})$ denotes the proximity between the center pixel and its neighborhoods. Next, the details of computing these probabilities are discussed.

The marginal probability $P(a_i)$ in Eqn. 3 represents the probability of label a_i in $\mathcal{G}$ which is the set of pixels in a rectangular region around the center pixel. During experiments, the size of the rectangular region is controlled by a user input radius R . Therefore, the statistics of pixels' labels in the region can be used to compute $P(a_i)$ as:

$$P(a_i) = \frac{1}{(2R+1)^2} \sum_{x'=x-R}^{x+R} \sum_{y'=y-R}^{y+R} |\mathcal{B}(x', y') \cap a_i|, \quad (4)$$

where $(2R+1)^2$ is the total number of pixels in the rectangular region, (x, y) represents the position of the center pixel and $\mathcal{B}(x', y')$ denotes the foreground mask as well as the label matrix corresponding to the pixels of the current frame, in which x', y' are the indices. A high value of $P(a_i)$ implies higher likelihood of the label a_i appearing in the region, and there is a high possibility that the label of the center pixel is a_i due to the correlation with their neighborhoods. In this condition, the label of the center pixel is correctly labelled as a_i by the marginal probability even though random noise is generated. The marginal probability $P(a_i)$ thus has the ability to isolate the influence of noise, and it is one of the main reasons motivating us to devise a statistical refinement model based on the Bayesian theorem.

In contrast to the marginal probability $P(a_i)$, the conditional probabilities $P(v|\mathcal{G})P(\mathcal{G}|a_i)$ tend to protect the border between the foreground and background, since it is based on the proximity among pixels. In particular, $\mathcal{G}$ represents the set of neighboring pixels, v is the feature of the center pixel and a_i denotes the label. Let us denote the pixels in $\mathcal{G}$ as $g_j \in \mathcal{G}$, and the procedure of capturing the conditional probability can be distributed by g_j as $\sum P(v|g_j)P(g_j|a_i)$. Due to the fact that the foreground and background are exclusive to each other, it is fair to assume that there is no correlation between pixels with different labels, and $P(g_j|a_i)$ becomes 0 corresponding to g_j whose label is not a_i . Based on this assumption, we divide $\mathcal{G}$ into two subsets G_i and $(G_i)^c$ according to the labels of pixels, where G_i is the set of pixels with the label a_i and $(G_i)^c$ is the complement. Then, the conditional probability

$P(v|\mathcal{G})P(\mathcal{G}|a_i)$ can be computed as:

$$\begin{aligned}
P(v|\mathcal{G})P(\mathcal{G}|a_i) &= \sum_{g_j \in \mathcal{G}} P(v|g_j)P(g_j|a_i) \\
&= \sum_{g_j \in G_i} P(v|g_j)P(g_j|a_i) + \sum_{g_j \in (G_i)^c} P(v|g_j)P(g_j|a_i) \\
&= \sum_{g_j \in G_i} P(v|g_j)P(g_j|a_i) \\
\therefore P(g_j|a_i) &= 0 \quad \forall g_j \in (G_i)^c.
\end{aligned} \tag{5}$$

In particular, $P(g_j|a_i)$ represents the probability of pixels in $\mathcal{G}$ under the condition that a_i is the label of pixels. It can be computed by a statistical procedure, which is:

$$P(g_j|a_i) = \frac{1}{||G_i||_0} = \left(\sum_{x'=x-R}^{x+R} \sum_{y'=y-R}^{y+R} |\mathcal{B}(x', y') \cap a_i| \right)^{-1}, \tag{6}$$

where $\mathcal{B}(x', y')$ is the label matrix introduced in Eqn. 4. $||G_i||_0$ is the l_0 norm of G_i , which is also the number of pixels in G_i . $P(v|g_j)$ represents the proximity between the center pixel v and its neighborhood g_j . In this work, the proximity is approximated by a distance function $\mathcal{L}()$, which is a measure of distance between v and g_j , and an activation function $\varphi()$, which convert the distance to proximity. Mathematically, it is shown as:

$$P(v|g_j) \sim \varphi(\mathcal{L}(v, g_j)). \tag{7}$$

Then, with the help of Eqn. 6 and Eqn. 7, the conditional probability $P(v|\mathcal{G})P(\mathcal{G}|a_i)$ in Eqn. 5 can be expressed as:

$$\begin{aligned}
P(v|\mathcal{G})P(\mathcal{G}|a_i) &= \sum_{g_j \in G_i} P(v|g_j)P(g_j|a_i) \\
&\sim \frac{1}{||G_i||_0} \sum_{g_j \in G_i} \varphi(\mathcal{L}(v, g_j))
\end{aligned} \tag{8}$$

With several attempts with different activation functions, the following formula of φ achieves the best performance with acceptable complexity of computation:

$$\varphi(d) = \frac{1}{d}. \tag{9}$$

Moreover, inspired by the extraction of super-pixels [64], the distance function $\mathcal{L}(v, c_i)$ is devised as an Euclidean distance of the vectors consisting of five entries $\{l, a, b, x, y\}$, where (l, a, b) is the LAB color space of pixels and (x, y) is the spatial location. Mathematically, it is shown as follows:

$$\mathcal{L}(v, c) = \sqrt{\left(\frac{d_c}{\mu_c}\right)^2 + \left(\frac{d_s}{\mu_s}\right)^2} \tag{10}$$

where d_c and d_s denote the Euclidean distance of the LAB color and the spatial location between v and g_j . μ_c and μ_s are the means of the color and spatial distance, which are used to combine these two components. Mathematically, they are computed as:

$$\mu_c = \frac{1}{||\mathcal{G}||_0} \sum_{g_j \in \mathcal{G}} d_c(g_j, v), \quad \mu_s = \frac{1}{||\mathcal{G}||_0} \sum_{g_j \in \mathcal{G}} d_s(g_j, v) \tag{11}$$

Now, we have the conditional probability, and the marginal probability. The label with the highest probability is used as the label of the center pixel located at (x, y) . Therefore, our Bayesian refinement model is described as:

$$\begin{aligned}
\mathcal{M}(x, y) &= \mathcal{F}(\mathcal{U}, \mathcal{B}) = \operatorname{argmax}_{a_i} P(a_i|\mathcal{U}(x, y)) \\
&= \operatorname{argmax}_{a_i} \frac{P(a_i)}{||G_i||_0} \sum_{g_j \in G_i} \varphi(\mathcal{L}(\mathcal{U}(x, y), g_j)) \\
&= \operatorname{argmax}_{a_i} \sum_{g_j \in G_i} \varphi(\mathcal{L}(\mathcal{U}(x, y), g_j)) \\
G_i &= \{\mathcal{U}(x', y') | \mathcal{B}(x', y') \cap a_i = 1\}
\end{aligned} \tag{12}$$

$$\begin{aligned}
\therefore \frac{P(a_i)}{||G_i||_0} &= (2R+1)^2 \\
&\& \operatorname{argmax}_x Nx = \operatorname{argmax}_x x \quad \forall N > 0
\end{aligned}$$

where $\mathcal{M}(x, y)$ is the output of the refinement model in which $x' \in [x-R, x+R]$ and $y' \in [y-R, y+R]$ are the indices of pixels belonging to G_i and (x, y) is the location of the center pixel, $\mathcal{B}(x', y')$ is the foreground mask generated by distribution learning and $\mathcal{U}(x, y) = v$ denotes the feature captured from the pixel located as (x, y) . $\mathcal{L}$ and φ are the distance function and the activation function discussed in Eqn. 10 and Eqn. 9.

The noise included in the input of the refinement model influences the output. Thus, our Bayesian refinement model cannot handle stubborn noise in one iteration. Fortunately, the output of the Bayesian refinement model is empirically better than the input, and can be used in the input again to generate better results iteratively, as shown in the right part of Fig. 4. Therefore, our Bayesian refinement model is devised as an iterative procedure as:

$$\mathcal{M}_n(x, y) = \mathcal{F}(\mathcal{U}(x, y), \mathcal{M}_{n-1}(x, y)), \tag{13}$$

where $\mathcal{M}_n(x, y)$ is the foreground mask in iteration n , $\mathcal{F}$ is the Bayesian refinement model and $\mathcal{U}$ is the set of features captured from pixels. Considering performance and computation cost, the number of iteration is set to 3 and the radius R is set to 4 during all experiments. In particular, With a larger iteration number, the performance of the proposed approach is supposed to be better, but the computational cost will increase. Moreover, the robustness of the refinement model is improved when R becomes larger, since more pixels are considered during refinement. However, a larger R increase the computational cost as well.

IV. EXPERIMENTS

In this section, comprehensive experiments for evaluating the proposed approach are presented. In Section IV-A, the transferability of our D-DPDL model is discussed, in which the training videos and testing videos are different. The efficiency of our dynamic training strategy and Bayesian refinement model, which are the two main differences compared with our previous work [1], is demonstrated in Section IV-B. Then, comprehensive comparison between the proposed

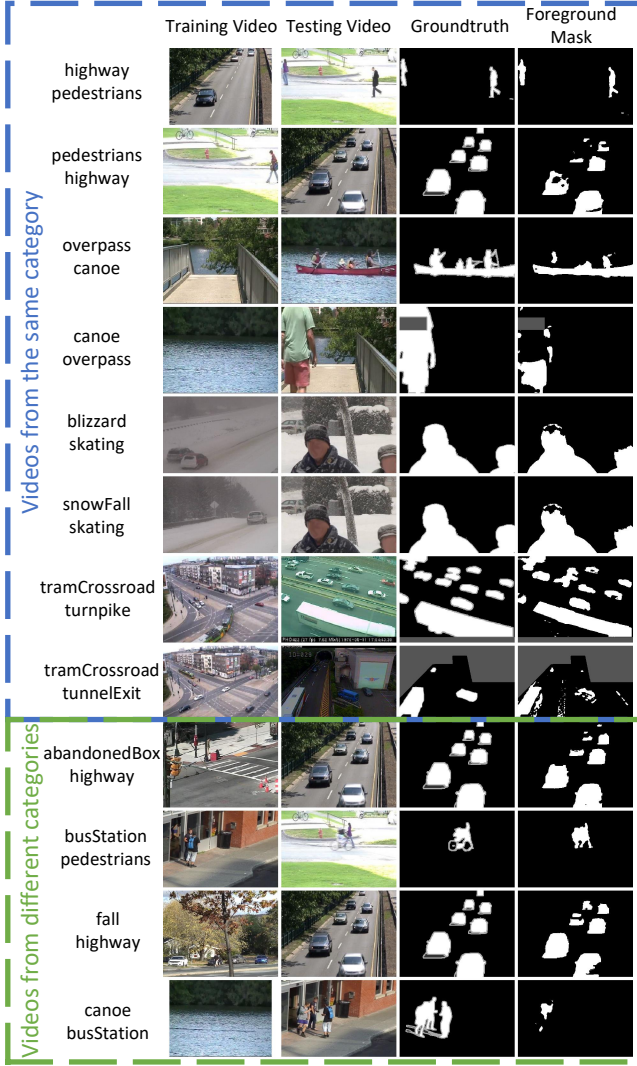


Fig. 5. The qualitative evaluation of the transferability of the proposed approach in several videos of CDnet2014 [65] dataset. In particular, the training frames and testing frames are obtained from different videos.

approach and several state-of-the-art methods including deep learning methods is presented in Section IV-C. Finally, a discussion of the computational cost of the proposed approach is proposed in Section IV-D.

A. Transferability of D-DPDL

In this section, the transferability of our D-DPDL algorithm is demonstrated. Previous approaches related to deep learning only guarantee the effectiveness under the assumption that the training frames and testing frames are obtained from the same scene. In contrast, our D-DPDL model is valid even when the testing frames are captured from a scene different from the one used for capturing training frames. This happens because the distribution of pixels, which are learned by the D-DPDL model, is a general feature regardless of the scene.

In order to demonstrate transferability, the proposed approach is evaluated for image frames which do not belong to the training video. During the experiments, several challenging

TABLE II
QUANTITATIVE TRANSFERABILITY EVALUATION OF THE PROPOSED APPROACH USING RE, PR AND FM METRICS, IN WHICH THE TRAINING AND TESTING VIDEOS ARE DIFFERENT.

Training Sets		Testing Sets		Performance		
Video	Category	Video	Category	Re	Pr	Fm
highway	baseline	pedestrians	baseline	(0.9723, 0.8138, 0.8860)		
pedestrians	baseline	highway	baseline	(0.6132, 0.9998, 0.7602)		
overpass	Dyn. Bg.	canoe	Dyn. Bg.	(0.8231, 0.9948, 0.9008)		
canoe	Dyn. Bg.	overpass	Dyn. Bg.	(0.3709, 0.9530, 0.5340)		
blizzard	Bad Wea.	skating	Bad Wea.	(0.8471, 0.9870, 0.9117)		
snowFall	Bad Wea.	skating	Bad Wea.	(0.7618, 0.9924, 0.8619)		
tramCrossroad	Low Fr.	turnpike	Low Fr.	(0.9188, 0.9537, 0.9359)		
tramCrossroad	Low Fr.	tunnelExit	Low Fr.	(0.7544, 0.3724, 0.4987)		
abandonedBox	Int. Mot.	highway	baseline	(0.8103, 0.9996, 0.8951)		
busStation	Shadow	pedestrians	baseline	(0.9747, 0.8668, 0.9176)		
fall	Dyn. Bg.	highway	baseline	(0.8102, 0.9994, 0.8949)		
canoe	Dyn. Bg.	busStation	Shadow	(0.0718, 0.9738, 0.1337)		
Average				(0.7274, 0.9089, 0.7609)		

videos are selected from Change Detection net 2014 (CD-net2014) dataset [65] for evaluation. In particular, 10 frames are extracted from the training video to learn the distribution information of pixels, and all frames of the testing video are utilized to evaluate the proposed approach. Quantitative and qualitative evaluations are shown in Table II and Fig. 5 respectively. The evaluations are divided into two parts. First, the training videos and testing videos are selected from the same category, which means the distribution of pixels from training videos and testing videos may share some common features. For example, most of the pixels on the background scene of videos in the category of “Dynamic Background” shared a repetitive pattern of variation in pixels’ observations. In the second part, the training video and testing video are selected from different categories to emphasize the difference between training videos and testing videos for evaluating the transferability of the proposed approach.

As shown in Table II and Fig. 5, our D-DPDL model achieves 0.7609 on the average *Fm* metric of all testing videos, when the scenes in the training and testing videos are different. Therefore, it is fair to claim that our D-DPDL is effective when the training and testing videos are completely different, demonstrating the transferability of the proposed approach. Moreover, the performance of the transferability is reasonably different in different cases due to the diversity of scenes in these videos. For example, in the first case “highway-pedestrians,” the proposed approach is trained on the video “highway” and tested on the video “pedestrians.” Since the scenes for these two videos do share several similar parts like the plants and roads, the proposed approach generates promising foreground results in this case. However, when we simply switch these two videos in the second case “pedestrians-highway,” the performance of the proposed approach has an obvious decline, due to the fact that the diversity of the scene with pedestrians is lower than the one with highways. The diversity of scenes thus plays an important role in the transferability of the proposed approach. This is more clear from the cases “overpass-canoe” and “canoe-overpass,” in which the video “overpass” includes water, plants and roads, but the video “canoe” only contains water and plants. When the proposed approach is trained on the video “overpass”

TABLE III

QUANTITATIVE DEMONSTRATION OF THE EFFECTIVENESS OF THE DYNAMIC TRAINING STRATEGY AND THE BAYESIAN REFINEMENT MODEL USING RE, PR AND FM METRICS. THE PERFORMANCES OF OUR D-DPDL WITH BAYESIAN REFINEMENT MODEL ARE SHOWN IN THE FIRST ROW, THE ONE WITHOUT ARE PRESENTED IN THE SECOND ROW AND THE THIRD ROW DEMONSTRATES THE PERFORMANCES OF OUR PREVIOUS DPDL METHOD, IN WHICH THE DYNAMIC TRAINING STRATEGY IS NOT USED, IN CDNET2014 [65] DATASET.

Videos		highway	office	sidewalk	boulevard	abandonedBox	parking	snowFall	wetSnow	bungalows	cubicle	Average
Bayesian Refinement	Re	0.9666	0.9498	0.8007	0.6404	0.7990	0.8690	0.7569	0.6011	0.8091	0.9010	0.8094
	Pr	0.9978	0.9104	0.6304	0.6610	0.9759	0.6524	0.3754	0.8259	0.9923	0.6611	0.7683
	Fm	0.9819	0.9297	0.7054	0.6505	0.8787	0.7453	0.5019	0.6958	0.8914	0.7626	0.7743
Dynamic Training	Re	0.9076	0.9275	0.6932	0.5906	0.7779	0.7641	0.7149	0.5435	0.7429	0.8224	0.7485
	Pr	0.9819	0.9061	0.3672	0.4222	0.8456	0.4211	0.2810	0.7082	0.9807	0.6230	0.6537
	Fm	0.9433	0.9166	0.4801	0.4924	0.8103	0.5430	0.4035	0.6150	0.8454	0.7090	0.6759
DPDL [1]	Re	0.9213	0.3351	0.6062	0.7327	0.8063	0.1180	0.7970	0.3772	0.9523	0.6880	0.6334
	Pr	0.9444	0.9900	0.2397	0.1490	0.6515	0.9471	0.1987	0.3175	0.2635	0.3984	0.5100
	Fm	0.9327	0.5007	0.3436	0.2477	0.7207	0.2098	0.3181	0.3448	0.4128	0.5046	0.4536



Fig. 6. Qualitative comparison between D-DPDL with the Bayesian refinement model, the one without it and our previous DPDL [1] method in CDnet2014 [65] dataset. In the third row, both the Bayesian refinement model and dynamic training strategy are used to generate the foreground masks of “Bayesian Refinement”, only dynamic training strategy are utilized to produce the foreground masks of “Dynamic Training” which is shown in the fourth row, none of these techniques are used in our previous DPDL [1] model in the last row.

and tested by the video “canoe,” it achieves 0.9008 for the Fm metric. On the other hand, the proposed approach only achieves 0.5340 when the training and testing videos are switched as shown in the case “canoe-overpass.” Hence, it only achieves 0.1337 when the proposed approach is applied to a more diverse video “busStation” as shown in the case “canoe-busStation,” in which the video “canoe” with low diversity is used for training and video “busStation” with high diversity is used for testing.

Similar phenomena are also shown in the cases “blizzard-skating” and “snowFall-skating,” in which the testing videos are the same but the training videos are different. As we can see in Fig. 5, the scenes in the video “blizzard” and “snowFall” are similar to each other. Thus, the proposed approach learns similar distribution information when these videos are used for training. Consequently, the proposed approach achieves similar performances when it is tested on the same video “skating” in these two cases. Similar conditions are also shown in the cases “tramCrossroad-turnpike” and “tramCrossroad-tunnelExit” where the proposed approach is trained on the same video but tested on different videos. In summary, the transferability of the proposed approach is highly dependant on the diversity of scenes included in the training video, where

more diverse distribution information can be learned by the proposed approach. When the diversity in the training video is higher than that of the testing video, the proposed approach generates good results. Otherwise, it fails like for the case “tramCrossroad-tunnelExit.”

In the second part of Table II and Fig. 5, the transferability of the proposed approach is demonstrated under the condition that the training videos and testing videos are captured from different categories, including the cases “abandonedBox-highway,” “busStation-pedestrians,” “fall-highway” and “canoe-busStation.” The proposed approach works well in the first three cases, as the diversity of scenes in the training videos is obviously higher than that of the testing videos. Therefore, the transferability of the proposed approach is affected by the diversity of scenes in the videos rather than their categories. Moreover, the proposed approach failed in the last case as the training videos only include plants and water, which do not appear in the testing videos. Since the objects in the training videos are not contained in the testing videos, the proposed approach cannot capture enough distribution information, which is the main reason for the poor performance in this case.

B. Bayesian Refinement Model and Dynamic Training

In this section, the validation of dynamic training and Bayesian refinement model is presented. The strategy of dynamic training is proposed to prevent the network from overfitting the pattern implied in random permutations. The Bayesian refinement model is devised to overcome the random noise generated by random permutations used in RPoTP features.

In order to present the validation of the Bayesian refinement model, we compare the performances of the proposed approach with or without the compensation of the Bayesian refinement model. As shown in Table III and Fig. 6, the performance of the proposed approach with refinement model is denoted by “Bayesian Refinement” and the performance without refinement model is denoted by “Dynamic Training.” Moreover, for validating the dynamic training strategy, the performance of our previous work DPDL [1] is presented as well for comparison, since the dynamic training strategy is not applied in DPDL [1]. During comparison, only one groundtruth frame is used for training and several challenging videos are selected from CDnet2014 dataset [65] for comparison.

As shown in Table III and Fig. 6, the dynamic training strategy and Bayesian refinement model have significant contribution to the efficiency and accuracy of the proposed approach. Compared to DPDL [1], the dynamic training strategy contributes over 30% improvement on the average Fm value, since RPoTP features are dynamically generated to prevent the network from overfitting to the pattern implied in random permutations. Then, the efficiency of the dynamic training strategy is demonstrated. In particular, the improvement resulting from the dynamic training strategy is more clear in several complex videos such as the video “office,” “bungalows” and “cubicle.” In these videos, the motion of moving objects is complex, where the network generates more noise in the foreground and background when it overfits to the pattern in a random permutation. For example, in the visual foreground shown in Fig. 6, the background of the video “office” has many randomly incorrect foreground points compared to the video “highway.”

The Bayesian refinement model further improves the performance of the proposed approach by handling the noise generated from the utilization of random permutations during capturing RPoTP features. This is shown in the visual foreground proposed in Fig. 6. The foreground mask produced by the dynamic training includes many noisy points in the background scenes as well as the foreground objects. By contrast, all these noisy points are removed in the foreground mask after refinement by the Bayesian refinement model. This is achieved by the marginal probability of the Bayesian refinement model discussed in Eqn. 4, which can handle random noise. Moreover, the conditional probability is utilized to protect the border between the foreground and background, and the Bayesian refinement model contributes an extra 10% improvement on the average Fm metrics. With the help of these two components, the proposed approach achieves much better performance with even less training frames. Specifically,

TABLE IV
QUANTITATIVE COMPARISON BETWEEN THE PROPOSED APPROACH AND STATE-OF-THE-ART USING FM METRIC ON LASIESTA [66] DATASET.

Videos	D-DPDL ₃	D-DPDL ₁	CueV2 [67]	Haines [8]	Cue [68]	SOBS [47]
I_SI_01	0.9596	0.8335	0.9208	0.9622	0.8143	0.9559
I_SI_02	0.8687	0.7963	0.8403	0.8130	0.7576	0.9409
I_CA_01	0.9309	0.6881	0.9062	0.9220	0.8424	0.8416
I_CA_02	0.8850	0.8724	0.7826	0.8656	0.6296	0.8731
I_OC_01	0.9710	0.8094	0.7013	0.8920	0.8274	0.9573
I_OC_02	0.9677	0.9331	0.8600	0.9526	0.8781	0.9508
I_IL_01	0.7161	0.6398	0.6452	0.8861	0.7966	0.1898
I_IL_02	0.8972	0.8749	0.6523	0.8122	0.7864	0.2312
I_MB_01	0.9699	0.9627	0.9543	0.9816	0.7779	0.9728
I_MB_02	0.9195	0.5937	0.9204	0.7064	0.6797	0.8517
I_BS_01	0.8371	0.5609	0.7132	0.6285	0.5065	0.4015
I_BS_02	0.6178	0.7393	0.6156	0.7333	0.6607	0.4021
O_CL_01	0.9792	0.9584	0.9508	0.6946	0.9280	0.9657
O_CL_02	0.9800	0.9819	0.9045	0.9588	0.8995	0.9760
O_RA_01	0.9072	0.4047	0.8453	0.8225	0.7462	0.8353
O_RA_02	0.9803	0.9812	0.8886	0.9590	0.8699	0.9591
O_SN_01	0.9690	0.4590	0.9317	0.3054	0.8214	0.9093
O_SN_02	0.9341	0.9421	0.6256	0.0426	0.0895	0.7116
O_SU_01	0.9065	0.8783	0.6774	0.8115	0.6527	0.8742
O_SU_02	0.9388	0.9171	0.7669	0.9021	0.8074	0.8843
Average	0.9068	0.7913	0.8051	0.7826	0.7386	0.7842

our D-DPDL model with 10 groundtruth frames for training achieves higher performance than our previous DPDL [1] model with 40 groundtruth frames for training.

C. Comprehensive Evaluation of D-DPDL

In this section, comprehensive comparisons between our Dynamic Deep Pixel Distribution Learning (D-DPDL) model and several sophisticated state-of-the-art methods including traditional methods and deep learning methods on LASIESTA [66] and CDnet2014 [65] datasets are presented. The results of the compared algorithms are directly referred from the implementation provided by authors.

It should be noted that the training data is important to supervised methods and has direct contribution to their performances. Methods with more training data are desired to achieve better performance. Especially, with the utilization of deep learning network, a few excellent methods [53]–[56] achieve almost perfect results in standardized datasets when many groundtruth foreground masks are used for training. For example, FgSegNet model proposed by Lim et al. [55] achieves over 98% F-scores when 200 groundtruth frames of each video are used for training. Though, these methods achieve better quantitative scores than the proposed approach, they also used much more groundtruth frames than our method requiring less than 10 groundtruth frames of each video. Therefore, in order to propose a fair comparison, besides unsupervised algorithms, several state-of-the-art methods claiming limited groundtruth frames are selected for comparison, which includes CwisarDH [45], DeepBS [60], GuidedBS [61], CNN-SFC [62] and our previous DPDL [1].

Among these supervised methods, CwisarDH [45] mentioned in their work that several pixel instances are taken in different frames of videos for training, but details are lacking. DeepBS [60] claims 5% of groundtruth frames of entire dataset are utilized for training, which means around $5\% \times 160000 = 8000$ groundtruth foreground masks are

TABLE V
QUANTITATIVE COMPARISON BETWEEN THE PROPOSED APPROACH AND STATE-OF-THE-ART USING FM METRIC ON CDNET2014 [65] DATASET.

Approach		Baseline	Dyn. Bg.	Cam. Jitt.	Int. Mot.	Shadow	Ther.	Bad Wea.	Low Fr.	Nig. Vid.	PTZ	Turbul.	Overall
unsupervised models	IUTIS-5 [13]	0.9567	0.8902	0.8332	0.7296	0.9084	0.8303	0.8248	0.7743	0.5290	0.4282	0.7836	0.7717
	SemanticBGS [69]	0.9604	0.9489	0.8388	0.7878	0.9478	0.8219	0.8260	0.7888	0.5014	0.5673	0.6921	0.7892
	AAPSA [11]	0.9183	0.6706	0.7207	0.5098	0.7953	0.7030	0.7742	0.4942	0.4161	0.3302	0.4643	0.6179
	MBS [31]	0.9287	0.7915	0.8367	0.7568	0.7968	0.8194	0.7980	0.6350	0.5158	0.5520	0.5858	0.7288
	PAWCS [34]	0.9397	0.8938	0.8137	0.7764	0.8913	0.8324	0.8152	0.6588	0.4152	0.4615	0.6450	0.7403
	ShareM [14]	0.9522	0.8222	0.8141	0.6727	0.8898	0.8319	0.8480	0.7286	0.5419	0.3860	0.7339	0.7474
	SuBSENSE [15]	0.9503	0.8177	0.8152	0.6569	0.8986	0.8171	0.8619	0.6445	0.5599	0.3476	0.7792	0.7408
	WeSamBE [32]	0.9413	0.7440	0.7976	0.7392	0.8999	0.7962	0.8608	0.6602	0.5929	0.3844	0.7737	0.7446
	GMM [4]	0.8245	0.6330	0.5969	0.5207	0.7370	0.6621	0.7380	0.5373	0.4097	0.1522	0.4663	0.5707
	RMoG [7]	0.7848	0.7352	0.7010	0.5431	0.7212	0.4788	0.6826	0.5312	0.4265	0.2470	0.4578	0.5735
	MSCL-FL [36]	0.9400	0.9000	0.8600	0.8400	0.8600	0.8600	0.8800	N/A	N/A	N/A	N/A	N/A
	DSPSS [70]	0.9664	0.9057	0.8662	0.7870	0.9177	0.7328	N/A	N/A	N/A	N/A	N/A	N/A
	STSHBM [71]	0.9534	0.9120	0.8503	0.8349	0.8930	0.8579	N/A	N/A	N/A	N/A	N/A	N/A
	DPGMM [8]	0.9286	0.8137	0.7477	0.5418	0.8127	0.8134	N/A	N/A	N/A	N/A	N/A	N/A
supervised models	DeepBS [60]	0.9580	0.8761	0.8990	0.6098	0.9304	0.7583	0.8301	0.6002	0.5835	0.3133	0.8455	0.7548
	GuidedBS [61]	0.9467	0.8266	0.8818	0.6229	0.8910	0.7490	0.8711	0.6396	0.5048	0.6057	0.8114	0.7591
	CNN-SFC [62]	0.9497	0.9035	0.8035	0.7499	0.9127	0.8494	0.9084	0.7808	0.6527	0.7280	0.8288	0.8243
	CwisarDH [45]	0.9145	0.8274	0.7886	0.5753	0.8581	0.7866	0.6837	0.6406	0.3735	0.3218	0.7227	0.6812
	DPDL ₁ [1]	0.7886	0.6566	0.5456	0.5115	0.6957	0.6697	0.6036	0.5966	0.3953	0.2942	0.6301	0.5807
	DPDL ₂₀ [1]	0.9620	0.8369	0.8627	0.8174	0.8763	0.8311	0.8107	0.6646	0.5866	0.4654	0.7173	0.7665
	DPDL ₄₀ [1]	0.9692	0.8692	0.8661	0.8759	0.9361	0.8379	0.8688	0.7078	0.6110	0.6087	0.7636	0.8106
	D-DPDL ₁	0.8964	0.8348	0.7731	0.6956	0.8284	0.8000	0.7313	0.6476	0.4735	0.3414	0.7708	0.7084
D-DPDL ₁₀	0.9743	0.9008	0.9098	0.8887	0.9526	0.8808	0.9001	0.7527	0.6505	0.6362	0.8073	0.8413	

required to train the network, where 160000 is the total number of groundtruth frames contained in CDnet2014 dataset [65]. Although GuidedBS [61] extracts no foreground masks from groundtruth frames for training, 200 foreground images of each video are assumed from the results generated by SubSENSE [15], which is a good background subtraction method. Therefore, we consider that $200 \times 53 = 10600$ foreground masks are required in GuidedBS [61]. This is around 6.625% of entire dataset. CNN-SFC [62] utilizes 4000 foreground masks from groundtruth data for training, which is around 2.5% of the entire dataset. For our previous DPDL model [1], DPDL₁, DPDL₂₀ and DPDL₄₀ utilize 1, 20 and 40 groundtruth masks for each video respectively, which are 0.033%, 0.66% and 1.32% of entire dataset. In contrast to the results of D-DPDL model proposed in this paper, 1 and 3 groundtruth frames are extracted for training in each video of LASIESTA [66] dataset to demonstrate the performance of D-DPDL₁ and D-DPDL₃ shown in Table IV, which takes around 0.24% and 0.72% of the groundtruth frames available in LASIESTA [66] dataset. In addition, in the CDnet [65] dataset, 1 and 10 frames are used for training in each video to demonstrate the performance of D-DPDL₁ and D-DPDL₁₀ shown in Table V and Fig. 7, which represents 0.033% and 0.33% of the groundtruth frames available in entire dataset.

Quantitative evaluation of the proposed approach on LASIESTA [66] dataset is shown in Table IV. The proposed approach achieves promising results even when only 1 groundtruth frame is used for training, and it generates the best performance when 3 groundtruth frames are used for training. There are 20 videos obtained from 10 different conditions in LASIESTA [66] dataset. Among them, videos “I_SI_01-02” are obtained from simple indoor scenes, in which the distribution of temporal pixels is easily classified, and the proposed approach achieves good performance. In addition, videos “I_CA_01-02,” “I_OC_01-02,” “I_IL_01-

02,” and “I_MB_01-02” include challenges of camouflage moving objects, occluded moving objects, moving objects with global illumination changes and modified background respectively. These challenges have significant influence on the distribution of pixels’ observations. Fortunately, benefiting from the strong learning ability of a deep learning network, our D-DPDL model handles these challenges and achieves promising results. In particular, D-DPDL₁ does not achieve good performance in video “I_IL_01,” since the illumination is changing during entire image sequences and 1 groundtruth frame may not contain enough information for the network to learn. When the number of groundtruth frames is increased to 3, the results of the proposed approach becomes promising.

However, in video “I_BS_02,” the performance of D-DPDL₃ is not good and it is even worse than their counterpart with 1 groundtruth frame for training. The reason is the imbalance between training instances for foreground and background. During the training of the proposed approach, the groundtruth frames are randomly selected from image sequences. When only 1 groundtruth frame is used for training, only the frames with foreground objects are used as candidates to make sure there are foreground instances in the training data. While for 3 groundtruth frames, the extra 2 groundtruth frames are randomly selected from the entire video, and it is possible that the selected groundtruth frames do not include any moving object to cause foreground and background instances being extremely imbalanced. This condition becomes even worse when the moving objects are relatively small and only appear in a few frames, which happens in the video “I_BS_02.” In contrast, videos “O_CL_01-02,” “O_RA_01-02,” “O_SN_01-02” and “O_SU_01-02” are obtained from outdoor scenes under different weather conditions, such as cloudy condition, rainy condition and so on. Our D-DPDL₃ achieves over 90% F-scores in all these videos, since the Bayesian refinement model has the ability to handle these

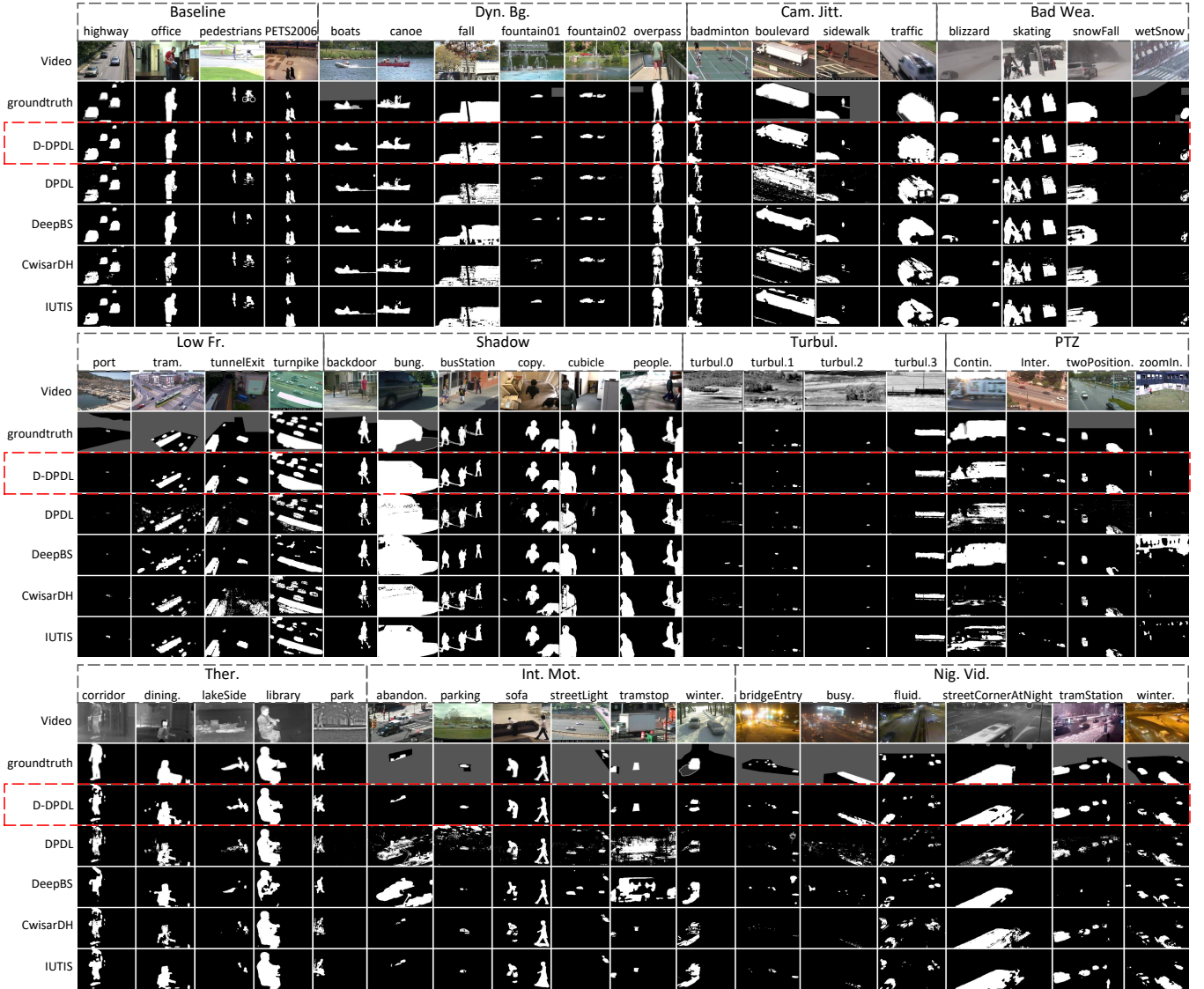


Fig. 7. Qualitative comparison between the proposed approach and several state-of-the-art methods including DeepBS [60], CwisarDH [45] and IUTIS-5 [13], on CDnet2014 [65] dataset, which contains 53 videos grouped into 11 categories.

isolated distractions from weather conditions.

Quantitative and qualitative evaluations of the proposed approach on CDnet2014 [65] dataset are shown in Table V and Fig. 7 respectively. To the best of our knowledge, CDnet2014 [65] dataset is the largest dataset for background subtraction, which contains 11 categories of videos. Therefore, the evaluation on CDnet2014 [65] dataset is fair and complete. As shown in Table V, our D-DPDL₁₀ has the best overall performance using 10 groundtruth frames compared to unsupervised methods as well as the supervised methods. Moreover, even with only one groundtruth frame, the proposed D-DPDL₁ still produces promising results which are close to the ones of MBS [31] or SuBSENSE [15]. Both of these demonstrate efficiency and accuracy of the proposed approach assuming limited groundtruth frames. In the “Baseline” category, the complexity of the distribution in pixel is not high, and the statistical information is easily learned by the proposed approach. The proposed D-DPDL₁₀ thus achieves excellent performance in

this category with 10 groundtruth frames used for training. It is higher than our previous DPDL₄₀ training by 40 groundtruth frames. This improvement demonstrates the effectiveness of our strategy of dynamic RPoTP feature generation, as well as the Bayesian refinement model. In addition, the improvement is more clear if we refer to the comparison between D-DPDL₁ and DPDL₁, where only one groundtruth frame is utilized for training. D-DPDL₁ achieves much higher performance compared to DPDL₁, since the dynamic training strategy is not applied in DPDL₁, the network tends to overfit to the pattern implied in random permutations when the training data is limited.

In the “Dynamic Background” category (Dyn. Bg.), the improvement of D-DPDL becomes more obvious due to the increasing complexity of pixel distribution compared to our previous DPDL model [1]. In a dynamic background, the observations of pixels have a pattern of repeating variation, which generates a multiple-peak distribution. Compared with

the DPDL model, the proposed D-DPDL model can focus better on the distribution with the help of the proposed dynamic training strategy. Therefore, our D-DPDL₁, which is trained with one groundtruth, even achieves performance close to DPDL₂₀ taking 20 groundtruth frames for training. However, due to the complexity of the distribution information in the pixels of a dynamic background, the network is trained by RPoTP features which unavoidably generates random noise during the utilization of random permutations. Thus, the Bayesian refinement model is proposed to handle the random noise, and the proposed D-DPDL model gives better results. This is more clear in the qualitative comparison shown in Fig. 7, in which the foreground masks generated by the D-DPDL model have almost no noise compared to our previous DPDL [1] model. Similarly, in the Camera Jitter (Cam. Jitt.) category, except for random permutations, the jitter of cameras also generates random noise. In this condition, the distribution of pixels' observation have multiple peaks as well. Benefiting from the learning ability of deep learning networks and our Bayesian refinement model's ability to handle random noise, the proposed approach generates promising results.

In the "Shadow" category, the main challenge comes from the illumination condition. The observations of pixels have significant variations when the illumination is occluded by moving objects. In this condition, the intensity of pixels' observations have a constant decline, which has a small influence on the waveform of their distribution. Previous methods, such as SemanticBGS [69], usually utilize texture features to overcome the illumination change. In contrast, the proposed D-DPDL model focuses on the distribution itself led by the illumination change, and has a strong ability to learn the distribution of pixels in the shadow or unshaded regions. Therefore, the proposed approach produces excellent foreground results in this category. A similar situation is also shown in the "Thermal" (Ther.) category. The "bad weather" (Bad Wea.) category includes several challenging videos. In particular, raining and snowing cause several hurdles in learning distributions. However, with the help of our Bayesian refinement model, the proposed approach still achieves promising results with 10 groundtruth frames for training compared to other deep learning methods such as CNN-SFC [62]. A similar condition also prevails in the "Low Frames" (Low Fr.) category, in which the different frame rates of cameras produce challenges to the methods related to distribution analysis such as GMM [4]. Fortunately, with the strong learning ability of our D-DPDL model, the proposed approach still generates promising results.

However, the proposed approach does not work well in the "Night Video" and "PTZ" category. It is understandable why the proposed approach cannot work in the PTZ category, since the videos in this category are obtained by moving cameras. For videos captured by a moving camera, the pixels no longer maintain their positions. The statistical information in the RPoTP feature is thus meaningless for learning the distribution. In the "Night Video" category, the proposed approach is still the best of two and close to CNN-SFC [62]. The distribution learning still works in this category, but the lighting in a video is too complex to be learned with limited groundtruth frames. Thus, the performance of the proposed

TABLE VI
THE RUNTIME OF DIFFERENT PARTS OF D-DPDL MODEL IN FRAMES
CAPTURED FROM SEVERAL VIDEOS WITH VARYING RESOLUTIONS
(SECONDS/FRAME).

Videos	Resolution	RPoTP Features Extraction	Foreground Segmentation	Bayesian Refinement	Overall
highway	320 × 240	0.5577	1.5453	3.3475	5.4505
fall	720 × 480	2.4810	10.0575	14.5516	27.0901
skating	540 × 360	1.3670	6.6319	7.2662	15.2651
traffic	320 × 240	0.5109	1.7026	2.7228	4.9363

approach is not that good.

D. Implementation details and Computational Cost

The D-DPDL model is implemented in Matlab with Mat-ConvNet [72], and the source code is available at https://github.com/zhaochenqiu/D_DPDL. The experiments are run on a PC with i7 CPU processor and GeForce GTX 1080 GPU processor in the Matlab 2018b environment under Ubuntu 16.04. In the evaluation experiments, all images have not been normalized. During training, 40 epochs are set as the maximum and the objective-epochs plots on four videos from different categories are shown in Fig. 8. Correspondingly, the processing time of the D-DPDL model on these videos are shown in Table VI.

As shown in Table VI, our D-DPDL is divided into three parts:

- 1) RPoTP features extraction: Observations of pixels are converted to RPoTP features and used as the input to the network.
- 2) Foreground segmentation: RPoTP features captured from pixels are classified by a convolutional neural network to generate foreground mask.
- 3) Bayesian refinement: The foreground mask generated from last step is iteratively refined by a Bayesian refinement model.

In total, the proposed D-DPDL model needs around 5 seconds to process an image frame with a resolution of 320 × 240, excluding the I/O time (i.e., reading frames from disk), since the proposed method is implemented purely in Matlab without any code optimization. In particular, the Bayesian refinement procedure uses most of the runtime, because it is an iterative procedure implemented in Matlab. In future, the refinement procedure will be re-implemented in C++, which will save considerable time. In addition, in the foreground segmentation part, we believe a multiple GPU implementations will accelerate the proposed approach. In the procedure of RPoTP features extraction, since loop computations are involved, the re-implementation by C++ will accelerate the proposed approach. Finally, the GPU implementation for random permutation should also accelerate the proposed approach.

V. CONCLUSION

In this paper we proposed the Dynamic Deep Pixel Distribution Learning (D-DPDL) model for background subtraction. In particular, Random Permutation of Temporal Pixels (RPoTP) features is proposed to indirectly force the convolutional neural

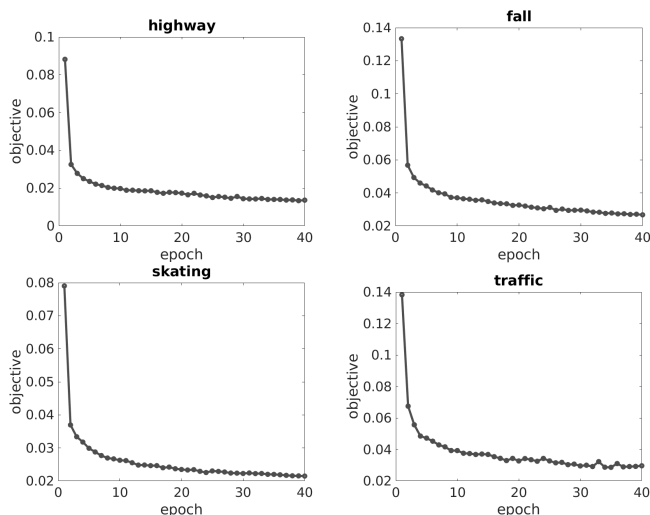


Fig. 8. The objective-epoch plots on four videos from different categories, in which video “highway” comes from category “Baseline,” video “fall” is extracted from category “Dynamic Background,” video “skating” is from category “Bad Weather” and video “traffic” belong to category “Camera Jitter.”

network to learn statistical distributions. Also, a Bayesian refinement model was proposed to handle the random noise generated during random permutations. Due to the pixel-wise representation of the RPoTP feature, a large number of RPoTP features can be captured with limited groundtruth frames for training the proposed approach. Moreover, since the statistical distribution is a general feature regardless of the scene, our D-DPDL model is effective even under the condition that the training videos and testing videos are completely different. Comprehensive evaluations comparing with other state-of-the-art methods demonstrate the superior efficiency of the proposed approach. In the future, we will work on optimizing the code and consider GPU implementation so that the proposed approach can get close to real-time performance.

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